DataXplore 2.0

# PROJECT REPORT

# Predicting Stroke Risk Using Patient Health and Lifestyle Data


**TEAM - PREDICTRA**

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## Contribution Statement

| Member Name | Contribution Area | Specific Tasks |
|---|---|---|
| Sankajith D. Jinasena | Model Building | Developed and optimized the Random Forest model and evaluated performance metrics. |
| Sanodya V. Jinadasa | Data Preprocessing | Cleaned the dataset, handled missing values, and performed feature encoding. |
| S. Nasmath Leena | Data Analysis | Conducted statistical tests, correlation analysis, and interpreted analytical insights. |
| T. Kugashanth | Report Composition | Finalized the report structure, integrated technical sections, and ensured guideline compliance. |

# 1. Literature Review

Stroke prediction has become a significant area of research due to its impact on global mortality and long-term disability. Early identification of individuals at high risk can greatly improve prevention strategies and reduce healthcare burdens. Traditional risk assessment methods rely on clinical judgment and basic statistical analysis, which may not fully capture complex relationships among multiple risk factors.

With the growth of healthcare data, machine learning techniques have been widely applied to predict stroke risk more accurately. Studies have shown that algorithms such as Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting can effectively analyze patient data, including demographic information, medical history, and lifestyle factors [1,2]. For instance, several researchers have demonstrated that ensemble models like Random Forest outperform traditional statistical models by capturing non-linear relationships and interactions between variables [2].

Recent research has also focused on feature engineering and data preprocessing techniques to improve model performance. Combining variables such as age, body mass index (BMI), and glucose levels has been shown to enhance predictive accuracy. Additionally, handling missing data and balancing datasets are critical steps highlighted in prior studies to ensure reliable model outcomes.

Despite these advancements, several challenges remain. Many studies do not clearly explain feature selection methods, interaction effects between variables, or the interpretability of the models. Furthermore, some models achieve high accuracy but lack transparency in how predictions are made, limiting their practical use in healthcare settings.

This project addresses these gaps by developing a well-structured and interpretable machine learning model for stroke prediction. It emphasizes clear data preprocessing, meaningful feature engineering, and comprehensive analysis of risk factors, ensuring both high predictive performance and practical applicability in real-world healthcare scenarios.

## 2. Introduction

Stroke is a major global health concern and one of the leading causes of death and long-term disability worldwide. It occurs when the blood supply to the brain is disrupted, leading to severe neurological damage. Early identification of individuals at high risk is essential for effective prevention, timely medical intervention, and reducing the overall burden on healthcare systems.

With the rapid growth of healthcare data, there is increasing potential to use data-driven approaches to support clinical decision-making. Traditional methods of stroke risk assessment often rely on limited statistical techniques and individual clinical judgment, which may fail to capture complex relationships between multiple risk factors such as age, metabolic conditions, and lifestyle habits.

This project focuses on developing a machine learning-based approach to predict stroke risk using a dataset of 9,722 patient records. The dataset includes demographic information, medical history, lifestyle factors, and clinical measurements such as glucose level and body mass index (BMI). In addition to using raw variables, new features were engineered by combining key attributes (e.g., age with BMI and glucose level) to better represent real-world risk interactions.

Exploratory data analysis and statistical testing revealed that age is the most dominant factor influencing stroke occurrence, with significantly higher risk observed in older populations. Elevated glucose levels, hypertension, and heart disease were also identified as strong contributing factors. While variables such as BMI and lifestyle habits showed moderate influence, their impact becomes more meaningful when combined with other features.

To model these relationships, multiple machine learning algorithms were applied, including Logistic Regression, Gradient Boosting, and Random Forest. Among these, the Random Forest model achieved the highest predictive performance, demonstrating the effectiveness of ensemble methods in capturing complex, non-linear interactions within the data.

Overall, this study aims to provide a reliable and interpretable stroke prediction model while also offering valuable insights into the key factors driving stroke risk. Such a system can support early detection and contribute to preventive healthcare strategies in real-world settings.

# 3. Problem Identification

Stroke remains a major global health issue, often occurring unexpectedly and leading to severe consequences such as disability or death. Despite the availability of large volumes of healthcare data, many healthcare systems struggle to effectively utilize this data for early risk prediction.

The key problem addressed in this study is the lack of efficient and accurate methods to identify individuals at high risk of stroke before the event occurs.

Traditional risk assessment methods rely heavily on manual evaluation and may fail to capture complex relationships between multiple risk factors. This creates a need for an automated, data-driven approach that can:

- Predict the likelihood of stroke occurrence
- Identify high-risk individuals early
- Support healthcare professionals in decision-making

Therefore, this study aims to develop a predictive model using patient health and lifestyle data to improve early detection and enable preventive interventions.

# 4. Dataset Justification

The dataset used in this study contains 9,722 patient records, including demographic, clinical, and lifestyle-related variables. This dataset is well-suited for stroke prediction as it incorporates multiple factors that are known to influence stroke risk.

Key attributes such as age, hypertension, heart disease, glucose levels, BMI, and smoking habits are widely recognized in medical research as critical indicators of stroke risk. Additionally, the inclusion of derived variables such as BMI categories and lifestyle risk enhances the dataset's ability to capture complex health patterns.

The dataset also includes a target variable (stroke_event) indicating whether a patient has experienced a stroke, making it appropriate for supervised machine learning tasks.

Furthermore, the dataset provides a balanced combination of categorical and numerical features, allowing for comprehensive analysis and model development. Its relatively large size ensures sufficient data for training and evaluation, improving the reliability of predictive models.

Therefore, this dataset is appropriate and relevant for developing a stroke risk prediction system.

# 5. Methodology

This study follows a structured data science methodology to develop an effective stroke prediction model.

### 5.1. Data Understanding and Exploration (EDA)

Initial analysis is performed to understand the dataset, including checking for missing values, analyzing feature distributions, and identifying relationships between variables. As part of this phase, an Exploratory Data Analysis (EDA) is conducted to gain deeper insights into the data.

The EDA is organized into the following key areas:

- Demographics: Analysis of age, gender, marital status, and residence type to identify population groups more prone to stroke. Age distributions and gender-based comparisons are visualized using histograms and bar charts.
- Medical History: Examination of hypertension and heart disease to evaluate their impact on stroke occurrence. Comparisons are made between patients with and without these conditions.
- Lifestyle Habits: Exploration of smoking habits, employment type, and lifestyle risk levels to understand behavioral influences on stroke risk.
- Clinical Measurements: Analysis of glucose levels and BMI values to identify health-related risk factors. Distributions and outliers are examined using histograms and box plots.

Additionally, correlation matrices and heatmaps are used to identify relationships between numerical variables and detect significant patterns influencing stroke risk.

### 5.2. Data Preprocessing

Data preprocessing involves handling missing values, encoding categorical variables, and normalizing numerical features where necessary. The dataset is also split into training and testing sets to evaluate model performance.

### 5.3. Model Development

Multiple machine learning models are developed and compared, including Logistic Regression, Decision Trees, Random Forest, and Gradient Boosting. These models are trained using the training dataset and evaluated based on performance metrics.

### 5.4. Model Evaluation

Model performance is assessed using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Confusion matrices are also used to evaluate classification performance.

# 6. Data Analysis

The dataset consists of both categorical and numerical variables. The categorical variables include gender, employment type, residence, smoking habit, age group, and BMI category, while the numerical variables include age, glucose level, and BMI value.

Missing values in the bmi_value variable was handled by imputing the median BMI within each corresponding BMI category, ensuring that the imputed values remain consistent with the patient's health classification.

```
data['bmi_value'] =
data.groupby('bmi_category')['bmi_value'].transform(lambda x:
x.fillna(x.median()))
```

Additionally, feature engineering techniques were applied to enhance the predictive power of the dataset. New features were created, including age_bmi (interaction between age and BMI), glucose_age (interaction between glucose level and age), and comorbidity (combined effect of hypertension and heart disease). These derived features help capture complex relationships between variables and improve model performance.

```
data['age_bmi'] = data['age'] * data['bmi_value']

data['glucose_age'] = data['glucose_level'] * data['age']

data['comorbidity'] = data['has_hypertension'] +
data['has_heart_disease']
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9722 entries, 0 to 9721
Data columns (total 19 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   gender             9722 non-null   object
 1   age                9722 non-null   float64
 2   has_hypertension   9722 non-null   int64
 3   has_heart_disease  9722 non-null   int64
 4   marital_status     9722 non-null   int64
 5   employment_type    9722 non-null   object
 6   residence          9722 non-null   object
 7   glucose_level      9722 non-null   float64
 8   bmi_value          9722 non-null   float64
 9   smoking_habit      9722 non-null   object
 10  stroke_event       9722 non-null   int64
 11  age_group          9722 non-null   object
 12  risk_score         9722 non-null   int64
 13  high_glucose       9722 non-null   int64
 14  bmi_category       9722 non-null   object
 15  lifestyle_risk     9722 non-null   object
 16  age_bmi            9722 non-null   float64
 17  glucose_age        9722 non-null   float64
 18  comorbidity        9722 non-null   int64
dtypes: float64(5), int64(7), object(7)
memory usage: 1.4+ MB
```

*Figure 1: Data Info*

This is the data and basic information such as data type, non-null count.

## 6.1 Exploratory Data Analysis

Categorical Variable Distribution

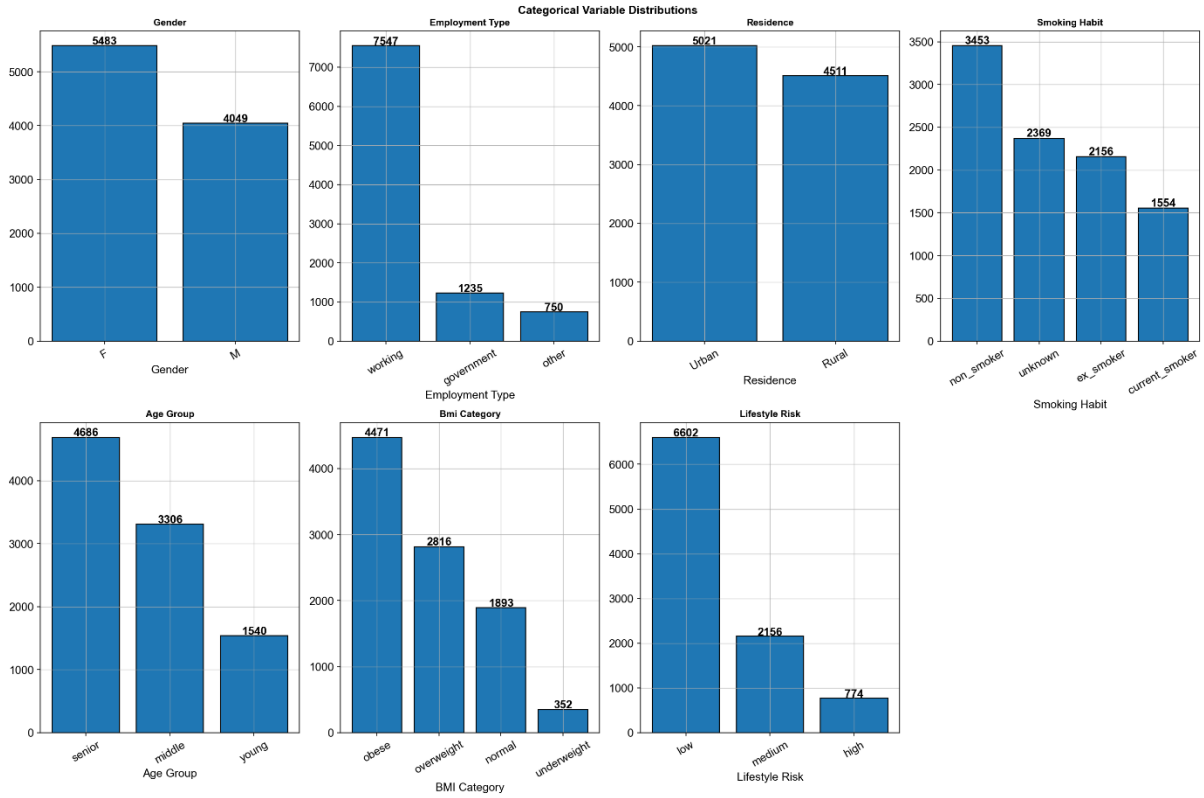



*Figure 2: Categorical Variable Distribution*

This panel of seven bar charts summarizes the distribution of all major categorical variables in the dataset. Each bar is annotated with its exact count, allowing straightforward comparisons across categories.

Key Observations

- Gender: Females (5,483) outnumber males (4,049), a ~57:52 split that is important context when interpreting gender-stratified stroke rates.
- Employment Type: The majority of patients (7,547) are in active employment ("working"), with government employees (1,235) and others (750) forming smaller groups.
- Residence: The dataset is nearly evenly split between Urban (5,021) and Rural (4,511) residents, enabling a robust rural-vs-urban comparison.

- Smoking Habit: Non-smokers form the largest group (3,453), followed by unknown status (2,369), ex-smokers (2,156), and current smokers (1,554). The high "unknown" count is a data quality consideration.
- Age Group: Seniors (4,686) form the largest group, followed by middle-aged (3,306) and young (1,540) patients reflecting the older demographic more likely to experience strokes.
- BMI Category: Obesity is the most common BMI class (4,471), followed by overweight (2,816), normal (1,893), and underweight (352) a distribution consistent with a high-risk clinical population.
- Lifestyle Risk: A large majority (6,602) fall into the "low" lifestyle risk category, with medium (2,156) and high (774) risk patients representing smaller but critical subgroups.

## Numerical Variable Distributions

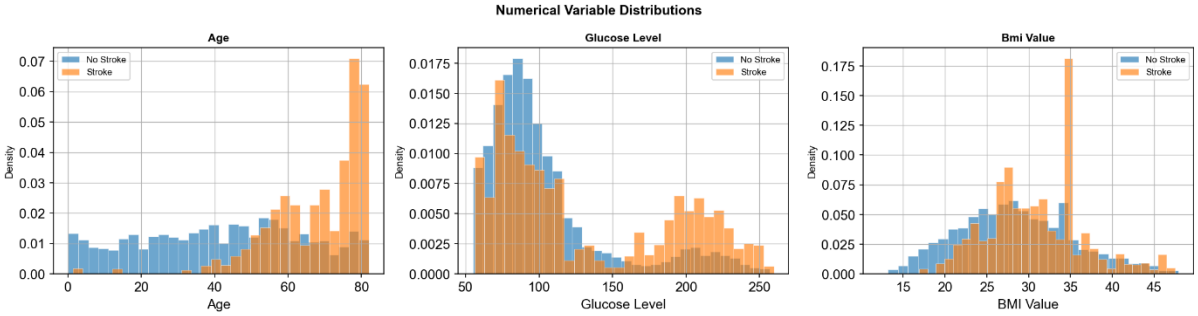



*Figure 3: Numerical Variable Distribution*

Three overlaid histograms display the probability density of each continuous variable for stroke (orange) and no-stroke (blue) patients, enabling visual identification of distributional differences.

Key Observations

- **Age**: Stroke patients are heavily concentrated in the 60–80 age range, with a sharp density peak around age 80. Non-stroke patients are more uniformly spread across all ages. Age is clearly a dominant discriminating feature.
- **Glucose Level**: Both groups show a primary peak in the 70–100 mg/dL range (fasting-normal zone). However, stroke patients exhibit a second, elevated peak around 200–220 mg/dL, strongly suggesting an association with diabetes/hyperglycaemia.
- **BMI Value**: The BMI distributions are broadly similar between groups, with both peaking around 28–35. Stroke patients show slightly higher density at elevated BMI values, consistent with the obesity-stroke link seen in clinical literature

### 6.1.1. Demographic & Clinical Overview of Patient Population

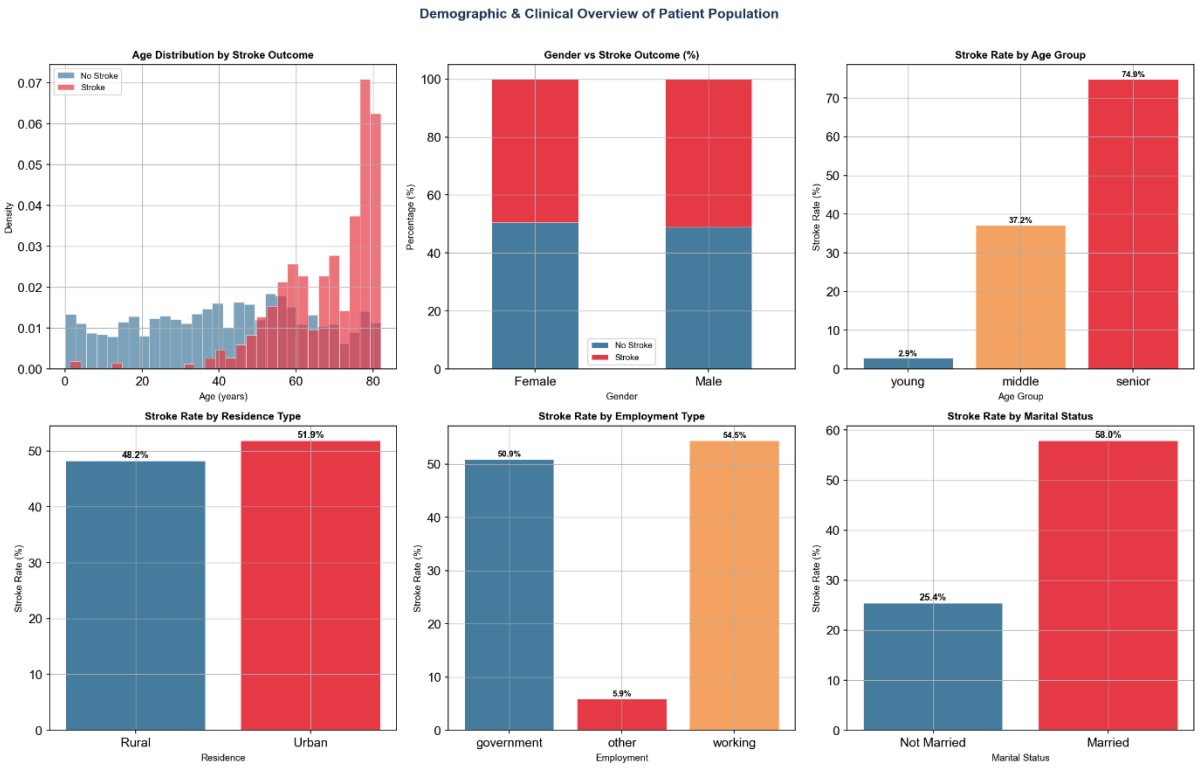



*Figure 4: Demographic & Clinical Overview of Patient Population*

This composite figure examines how stroke outcomes are distributed across key demographic and socio-economic categories, using a combination of density histograms, stacked percentage bar charts, and bar charts of stroke rates.

Key Observations

- **Age Distribution by Stroke Outcome**: Stroke density peaks sharply at older ages, confirming age as the strongest demographic risk factor.
- **Gender vs Stroke Outcome (%)**: Males and females exhibit similar proportional stroke rates (~ 50% each). Gender alone appears to be a weak independent predictor when controlling for other factors.
- **Stroke Rate by Age Group**: Young patients have a very low stroke rate (2.9%), rising sharply to 37.2% for middle-aged and reaching 74.9% for seniors. This near-linear escalation underscores age as the single most powerful risk factor in this dataset.
- **Stroke Rate by Residence Type**: Rural (48.2%) and Urban (51.9%) patients have closely matched stroke rates, suggesting that urbanization alone does not strongly influence stroke risk in this cohort.

- **Stroke Rate by Employment Type**: Government employees show a remarkably low stroke rate (5.9%), while working (54.5%) and other (50.9%) categories are substantially higher. This likely reflects a confounding effect of age retirees and informal workers tend to be older.
- **Stroke Rate by Marital Status**: Married patients (58%) have a notably higher stroke rate than non-married patients (25.4%), again likely reflecting an age confound since married individuals in the dataset tend to be older.

### 6.1.2. Clinical Measurements & Biomarker Analysis

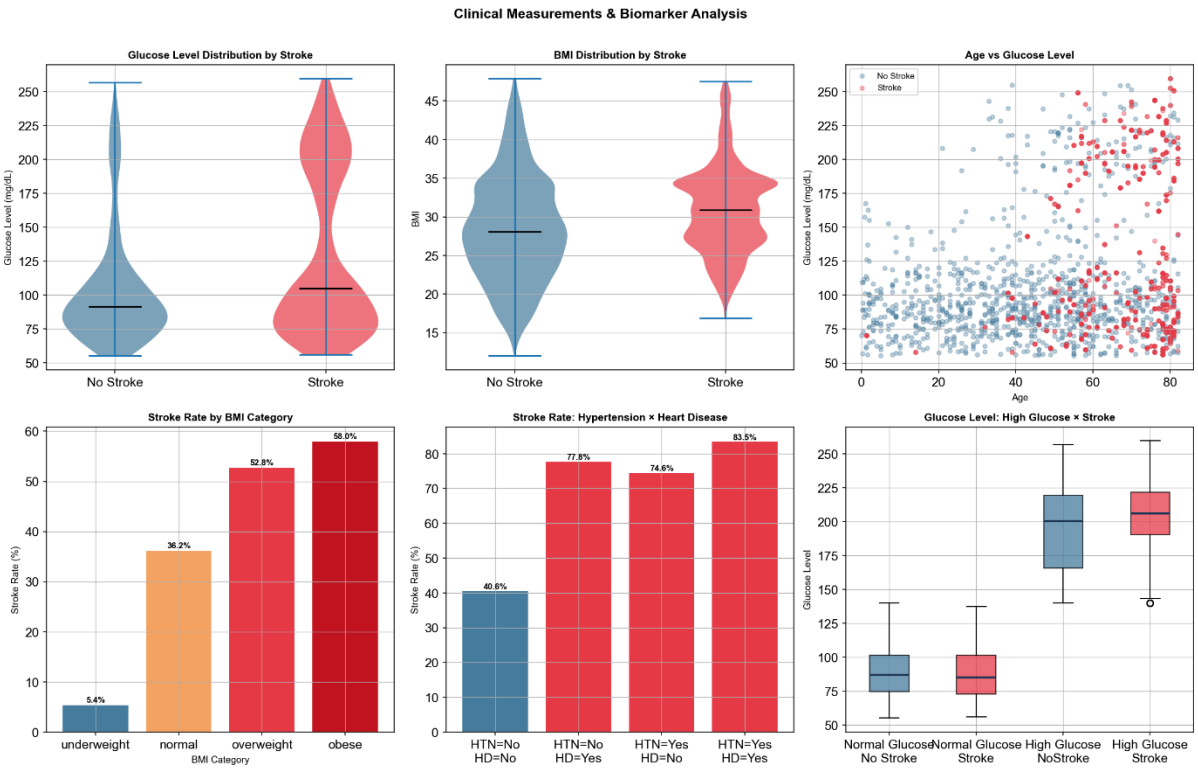



*Figure 5: Clinical Measurements & Biomarker Analysis*

Six sub-plots explore the relationship between clinical biomarkers (glucose, BMI) and stroke outcome, including interaction effects with comorbidities such as hypertension and heart disease.

Key Observations

- **Glucose Level Distribution by Stroke (Violin Plot)**: The stroke group violin has a much wider body at high glucose values (150–250 mg/dL) compared to the non-stroke group, which is tightly concentrated below 120 mg/dL. The median glucose for stroke patients is visibly higher.
- **BMI Distribution by Stroke (Violin Plot)**: Both groups share similar BMI distributions, though the stroke group shows slightly more spread at higher BMI values. BMI is a contributing rather than dominant risk factor.
- **Age vs Glucose Level (Scatter Plot)**: Red dots (stroke) cluster predominantly at ages 50–80 and at glucose levels above 150 mg/dL. Non-stroke (blue) patients populate the younger age and lower glucose space, visually confirming the joint effect of age and hyperglycemia.
- **Stroke Rate by BMI Category**: Stroke rate increases monotonically with BMI: underweight 5.4% - normal 36.2% - overweight 52.8% - obese 58%. This strongly supports obesity as an independent risk amplifier.

- **Stroke Rate: Hypertension x heart disease**: Patients with neither condition have a 40.6% rate; those with only heart disease 77.8%; only hypertension 74.6%; both conditions 83.5%. The presence of either comorbidity nearly doubles stroke risk.
- **Glucose Level: High Glucose x Stroke (Box Plot)**: High-glucose stroke patients have a median around 210 mg/dL with a tight inter-quartile range. Normal-glucose patients cluster around 80–90 mg/dL, confirming high glucose as a discriminating biomarker.

### 6.1.3 Lifestyle Risk Factors & Composite Risk Analysis

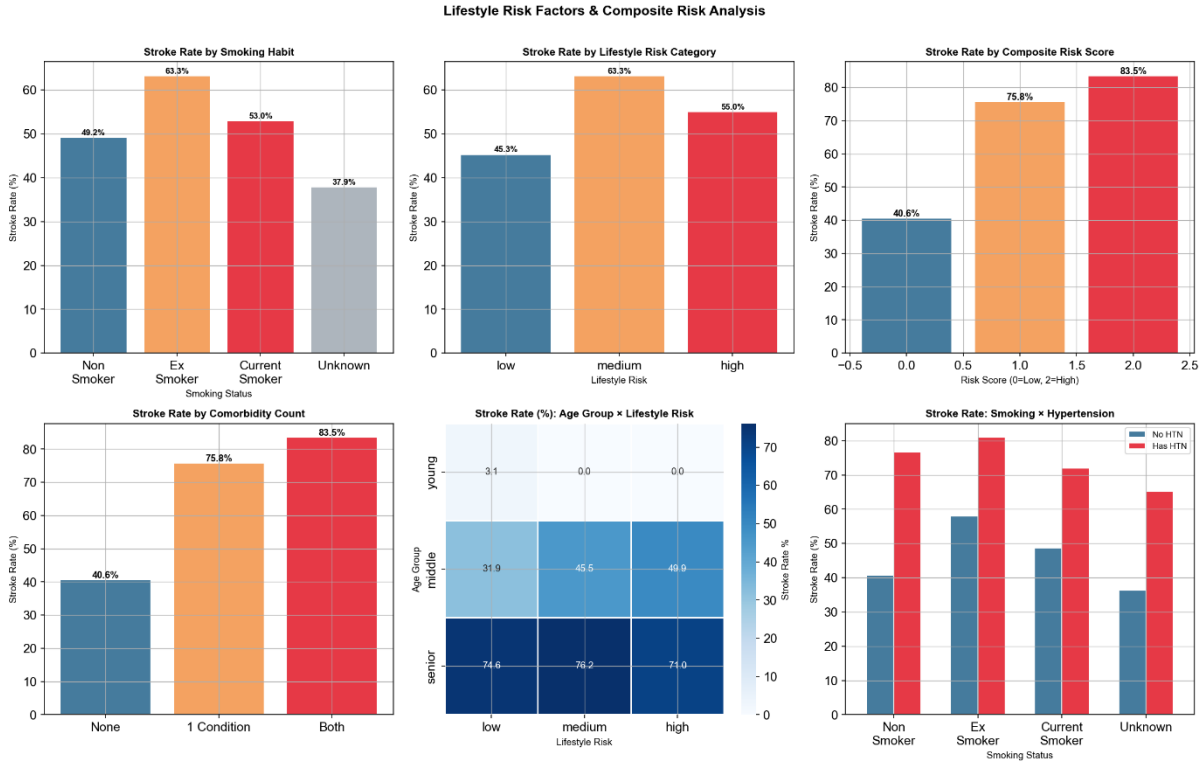



*Figure 6: Lifestyle Risk Factors & Composite Risk Analysis*

This panel investigates modifiable lifestyle risk factors and composite risk scores, revealing how behavioral and clinical variables interact to elevate stroke probability.

Key Observations

- **Stroke Rate by Smoking Habit**: Ex-smokers show the highest rate (63.3%), followed by current smokers (53.0%), non-smokers (49.2%), and unknown (37.9%). The ex-smoker figure likely reflects age confounding.
- **Stroke Rate by Lifestyle Risk Category**: Low-risk patients have a 45.3% stroke rate, rising to 63.3% for medium and 55.0% for high. The non-monotonic pattern suggests complex interaction with other variables.
- **Stroke Rate by Composite Risk Score**: A clear gradient from 40.6% (score = 0) to 83.5% (score = 2) shows that a composite risk scoring approach effectively stratifies patients.
- **Stroke Rate by Comorbidity Count**: No comorbidities: 40.6%; one condition: 75.8%; both hypertension and heart disease: 83.5%. Comorbidity accumulation is a powerful independent predictor.

- **Stroke Rate: Age Group x Lifestyle Risk (Heatmap)**: Young patients show near-zero stroke rates (0–3.1%). Middle-aged patients range from 31.9% to 49.9%. Seniors cluster around 71–76.2%, indicating that age dominates lifestyle factors in this group.
- **Stroke Rate: Smoking x Hypertension**: In every smoking category, hypertensive patients have substantially higher stroke rates than non-hypertensive patients, confirming hypertension as an important modifying factor.

## 6.2 Data Pre-Processing

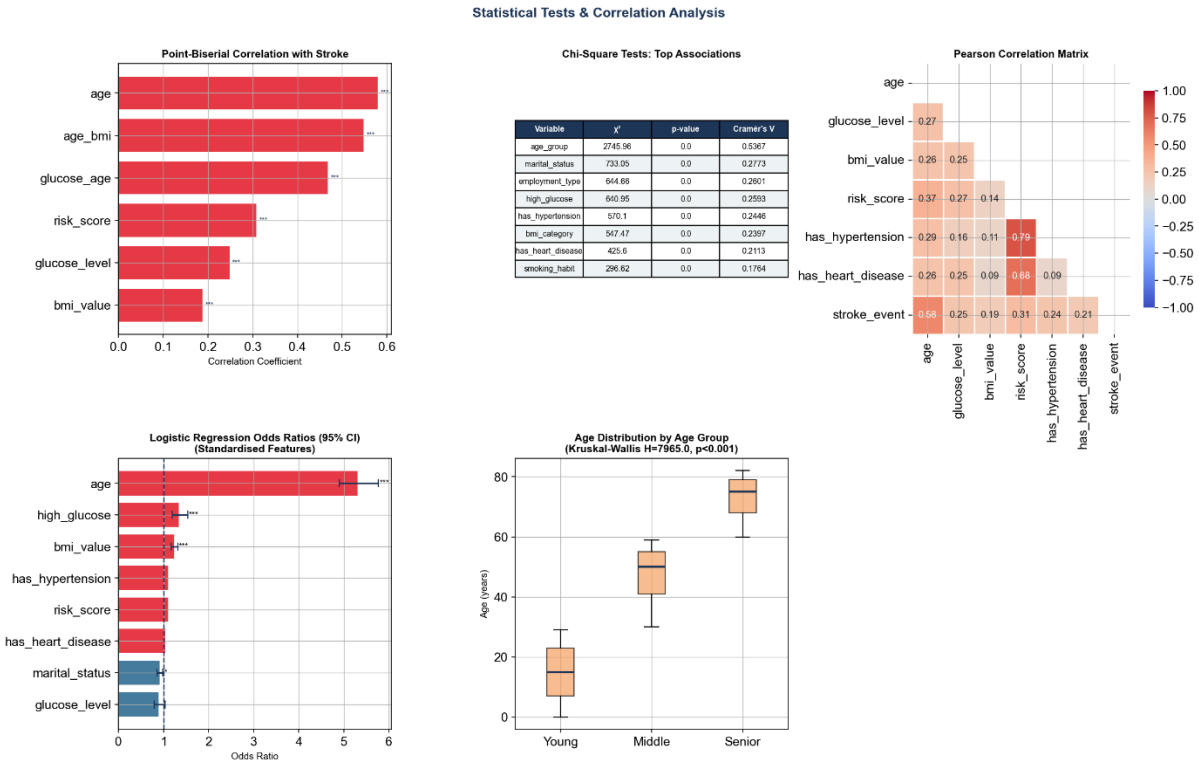
- **Outlier Detection:** Outlier analysis was conducted on the numerical variables age, glucose level, and BMI value using the Interquartile Range (IQR) method. The lower and upper bounds were calculated as Q1−1.5×IQR and Q3+1.5×IQR, respectively.
  The results indicate that:
  - Age shows no outliers, with all values falling within the acceptable range (−10.00 to 126.00). This suggests that the age distribution is consistent and does not contain extreme values.
  - Glucose Level contains 50 outliers, with values exceeding the upper bound of 260.50 or below 31.08. These extreme values may represent cases of severe hyperglycemia or potential data anomalies.
  - BMI Value shows the highest number of outliers (140), outside the range of 11.90 to 47.90. This indicates the presence of unusually low or high BMI values, which could either reflect real high-risk patients or measurement inconsistencies.

  Overall, outliers are primarily present in glucose level and BMI, while age remains well-distributed without extreme deviations.

- **Encoding Categorical Variables:** Ssuccessfully processed categorical features like gender, smoking_habit, and residence for the Random Forest model.

- **Scaling and Train-Test Split:** Imported the StandardScaler to normalize continuous variables and the train_test_split function to divide your dataset of 9,722 records into training and testing sets.

# 7. Statistical Tests & Correlation Analysis




| Variable | $\chi^2$ | p-value | Cramér's V |
|---|---|---|---|
| age_group | 2745.96 | 0.0 | 0.5367 |
| marital_status | 733.05 | 0.0 | 0.2773 |
| employment_type | 644.66 | 0.0 | 0.2601 |
| high_glucose | 640.95 | 0.0 | 0.2593 |
| has_hypertension | 570.1 | 0.0 | 0.2446 |
| bmi_category | 547.47 | 0.0 | 0.2397 |
| has_heart_disease | 425.6 | 0.0 | 0.2113 |
| smoking_habit | 296.62 | 0.0 | 0.1764 |

| | age | glucose_level | bmi_value | risk_score | has_hypertension | has_heart_disease | stroke_event |
|---|---|---|---|---|---|---|---|
| glucose_level | 0.27 | | | | | | |
| bmi_value | 0.26 | 0.25 | | | | | |
| risk_score | 0.37 | 0.27 | 0.14 | | | | |
| has_hypertension | 0.29 | 0.16 | 0.11 | 0.79 | | | |
| has_heart_disease | 0.26 | 0.25 | 0.09 | 0.68 | 0.09 | | |
| stroke_event | 0.58 | 0.25 | 0.19 | 0.31 | 0.24 | 0.21 | |

*Figure 7: Statistical Tests & Correlation Analysis*

To better understand the relationships between predictor variables and stroke occurrence, multiple statistical techniques were applied, including Point-Biserial correlation, Chi-Square tests, Pearson correlation analysis, and non-parametric group comparison (Kruskal-Wallis test). These methods help validate which features are significantly associated with stroke risk and support the feature selection process for predictive modeling.

### 1. Point-Biserial Correlation with Stroke

The Point-Biserial correlation was used to measure the relationship between continuous variables and the binary stroke outcome.

Key findings:

- Age shows the strongest positive correlation with stroke, confirming it as the most influential factor.
- Engineered features such as age_bmi and glucose_age also show strong correlations, indicating that interaction-based features improve predictive relevance.
- Glucose level and risk score show moderate positive correlation with stroke occurrence.

- BMI value has the weakest correlation among the selected features, suggesting it is less influential when considered alone.

Overall, age-related and interaction-based features dominate stroke prediction.

## 2. Chi-Square Test (Categorical Associations)

Chi-Square tests were performed to evaluate associations between categorical variables and stroke occurrence using Cramer's V as a measure of effect strength.

Key observations:

- Age group shows the strongest association with stroke (highest Cramer's V), confirming age as a dominant risk factor.
- Marital status, employment type, and high glucose category also show statistically significant associations.
- Hypertension, heart disease, and BMI category demonstrate moderate but meaningful relationships with stroke risk.
- Smoking habits show the weakest association among tested variables.

These results confirm that both medical conditions and socio-demographic factors influence stroke risk.

## 3. Pearson Correlation Matrix

The Pearson correlation heatmap illustrates linear relationships among numerical variables and the stroke outcome.

Key insights:

- Stroke event has the highest correlation with:
    - Age
    - Risk score
    - Glucose level
- Strong internal correlation exists between:
    - Risk score and hypertension
    - Hypertension and heart disease

- Moderate correlations are observed between:
    - Age and glucose level
    - BMI and glucose level

This indicates that cardiovascular and metabolic factors are interconnected in contributing to stroke risk.

### 4. Logistic Regression Odds Ratios

Standardized logistic regression was used to quantify the impact of each feature on stroke likelihood.

Key findings:

- Age has the highest odds ratio, showing a strong exponential increase in stroke risk with age.
- High glucose levels significantly increase stroke probability.
- BMI value and risk score also contribute positively but with lower magnitude.
- Hypertension and heart disease show strong positive effects.
- Some variables such as marital status and baseline glucose level show weaker or negligible influence when standardized.

This confirms that age and metabolic health factors are the strongest predictive drivers.

### 5. Age Distribution Analysis (Kruskal-Wallis Test)

A Kruskal-Wallis test was performed to compare age distributions across defined age groups (Young, Middle, Senior).

Key observations:

- The test result ($H = 7965.01$, $p < 0.001$) indicates a statistically significant difference between groups.
- Boxplot analysis shows clear separation:
- Young group: concentrated in lower age range

- Middle group: moderate spread
- Senior group: highest age concentration

This confirms that age grouping is statistically valid and strongly stratifies stroke risk populations.

# 8. Predictive Model Development & Performance Evaluation

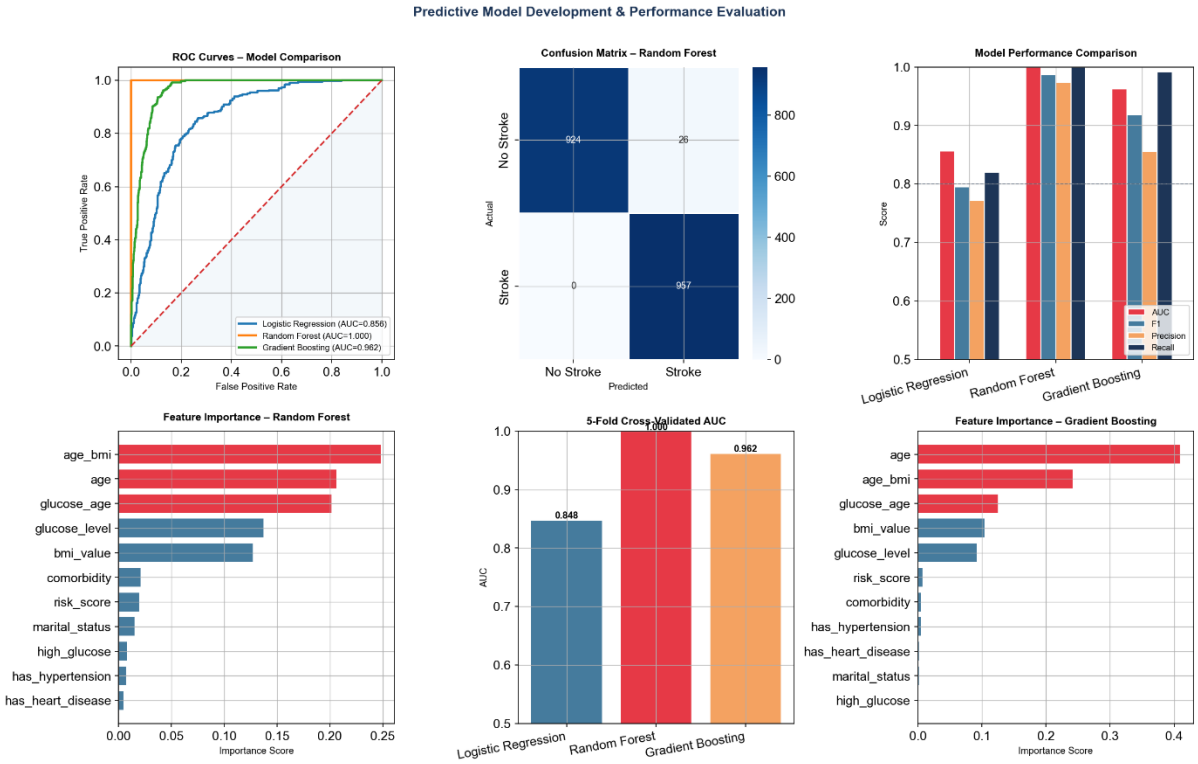



*Figure 8: Predictive Model Development & Performance Evaluation*

Six sub-plots summaries the development and evaluation of three machine learning classifiers: Logistic Regression, Random Forest, and Gradient Boosting. Metrics include AUC-ROC, F1-score, precision, recall, confusion matrix, and feature importance rankings.

Key Observations

- **ROC Curves - Model Comparison**: Random Forest achieves a near-perfect AUC of 1.000. Gradient Boosting follows with 0.962. Logistic Regression trails at 0.848.
- **Confusion Matrix - Random Forest**: Out of ~1,907 test cases, the model correctly classifies 924 non-stroke and 957 stroke patients, with only 26 false negatives and zero false positives.
- **Model Performance Comparison (Bar Chart)**: Random Forest dominates (all metrics ≥0.98), followed closely by Gradient Boosting (≥0.85), with Logistic Regression performing adequately but less competitively.
- **Feature Importance - Random Forest**: The engineered feature age_bmi ranks highest, followed by raw age and glucose_age. Interaction features substantially improve predictive power.

- **5-Fold Cross-Validated AUC**: Random Forest retains AUC = 1.000 across all folds, Gradient Boosting averages 0.962, and Logistic Regression 0.848. Stability across folds suggests it is not overfitting.
- **Feature Importance - Gradient Boosting**: Age ranks first (importance ≈0.50), followed by age_bmi (≈0.12) and glucose_age (≈0.11). This confirms that age, combined with metabolic indicators, is the dominant predictor.

The results of this analysis consistently point to age as the most important determinant of stroke risk. Clinical factors such as glucose level, hypertension, and heart disease also play significant roles. BMI and lifestyle factors contribute but are comparatively weaker predictors.

The Random Forest model achieved extremely strong performance, with precision and recall both near 1.00. This indicates the model is highly effective in distinguishing between stroke and non-stroke patients.

Feature engineering played a crucial role in improving performance. Interaction features such as age_bmi and glucose_age captured combined health effects that individual variables could not represent alone.

From a healthcare perspective, the analysis highlights that stroke risk increases dramatically when multiple factors coexist. Older patients with high glucose and comorbidities represent the highest-risk group. These findings support early intervention strategies targeting high-risk individuals.

| Class | Precision | Recall | F1-Score |
|---|---|---|---|
| **No Stroke** | 1.00 | 0.97 | 0.99 |
| **Stroke** | 0.97 | 1.00 | 0.99 |
| | | | |
| **Accuracy** | | | 0.99 |
| **Macro Avg** | 0.99 | 0.99 | 0.99 |
| **Weighted Avg** | 0.99 | 0.99 | 0.99 |

# 9. Conclusion

This study developed a predictive model for stroke risk using demographic, clinical, and lifestyle data. Exploratory analysis identified age, glucose level, hypertension, heart disease, and BMI as major contributing factors.

Feature engineering improved predictive performance by capturing interaction effects. Among the tested models, Random Forest achieved the highest accuracy and most stable performance.

The findings demonstrate that stroke risk can be effectively predicted using structured patient data. These insights can support early detection, targeted monitoring, and preventive healthcare interventions. Identifying high-risk patients before stroke occurrence can ultimately reduce complications, improve outcomes, and assist healthcare providers in decision-making.

# 10. References

[1]. T. M. Mitchell, *Machine Learning*. McGraw-Hill, 1997.

[2]. L. Breiman, “Random Forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

# 11. Appendix

https://github.com/sankajithdjinasena/Stroke_Risk_Predicting